13 For any three consecutive whole numbers, prove algebraically that

the largest number and the smallest number are factors of the number that is one less than the square of the middle number.

Three consecutive whole numbers: n, $n+1$, $n+2$.

1 mark
Consecutive integers

Middle squared, less 1: $(n+1)^2 - 1$

$= n^2 + 2n + 1 - 1 = n(n+2)$.

1 mark
Factorise square−1

This is the product of the smallest and largest.

So both are factors, as required.

1 mark
Conclude from identity

(Total for Question 13 is 3 marks)